

mutual inconsistency

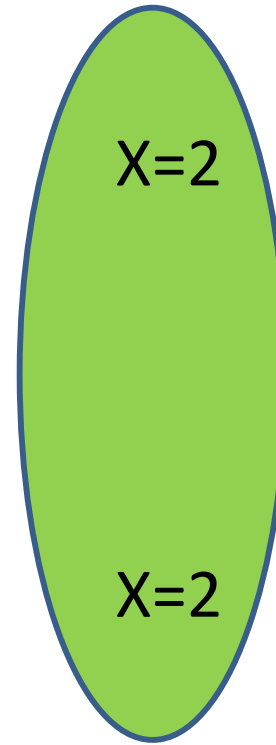
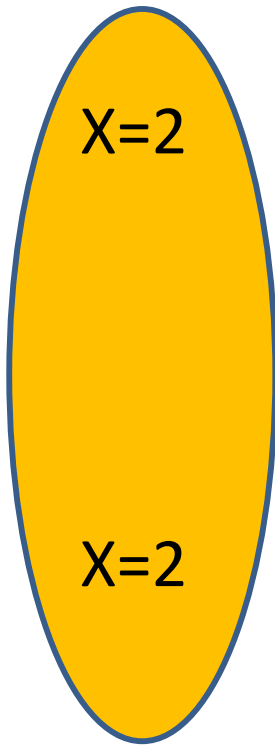
- Due to the possible breakdown of the communication links, the system can be divided into several **sub-systems** (or sub-networks).
 - There are **no communication** links between these sub-systems.
 - Each sub-system may contain one or several **copies** of a single object.
 - Objects are replicated.
- When the failed communication links are **repaired**, the sub-systems (or sub-networks) will merge.
- If the copies of the same object in different sub-systems can be modified during the breakdown, then a **consistent value** of the object might **not be found** when the sub-systems merge.
- The inconsistency arises from this situation is termed as **mutual inconsistency**.

$$X=2$$

$$X=2$$

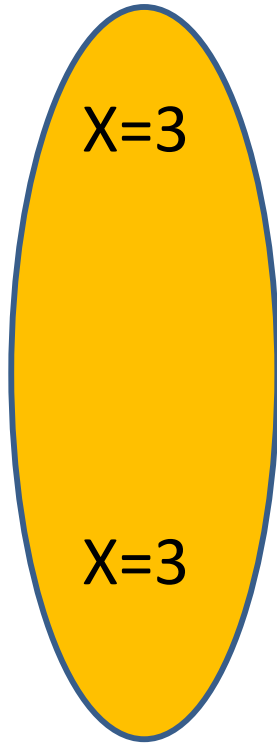
$$X=2$$

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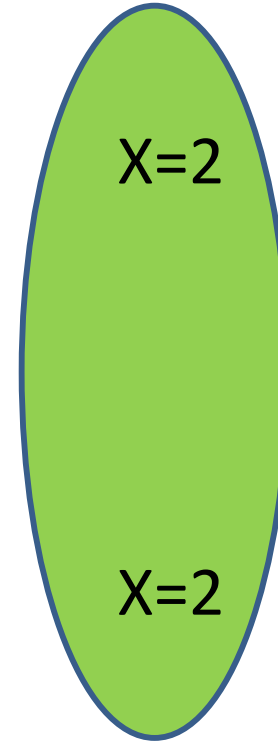


unconnected to network.

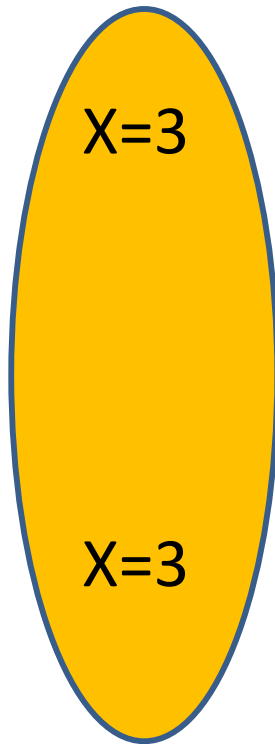
Update $X=3$



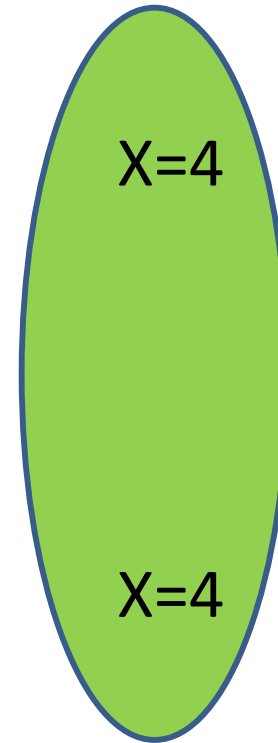
传播
propagate
→



Update X=3



Update X=4



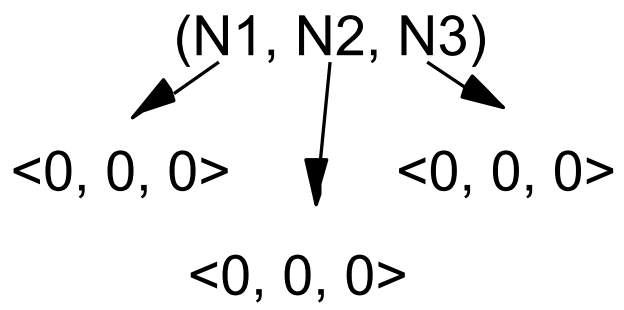
which to
use?

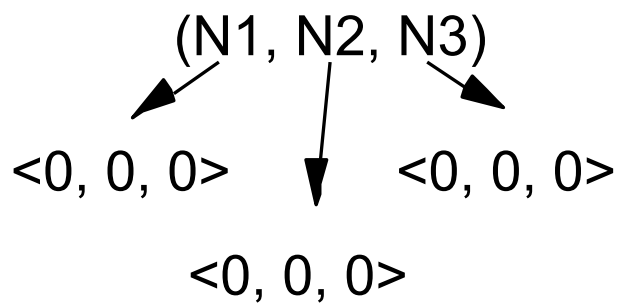
how many time be update.
↑
how many

$\langle a, b, c \rangle$ machine. $\Rightarrow$ how many value

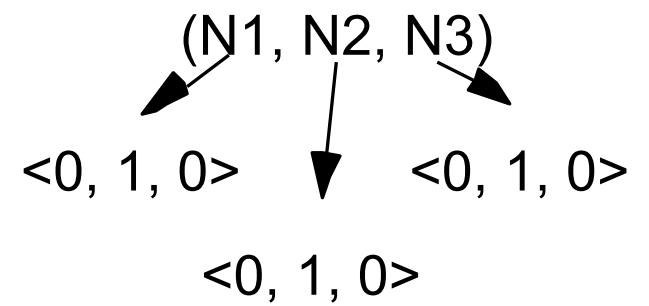
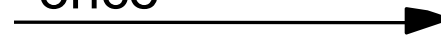
Detection of Mutual Inconsistency

- In the algorithm, a **vector** is associated with each copy of the replicated data.
 - **Each item** in the vector corresponds to a node in the system.
 - The item is used to represent the number of times that a copy on that site has been updated.
- It can be assumed that, for a particular **sub-system**, the **updates** to the copy of an object on one node are made to known to all the nodes in the system.
 - The **vectors** for the same object in one sub-system are always the same.

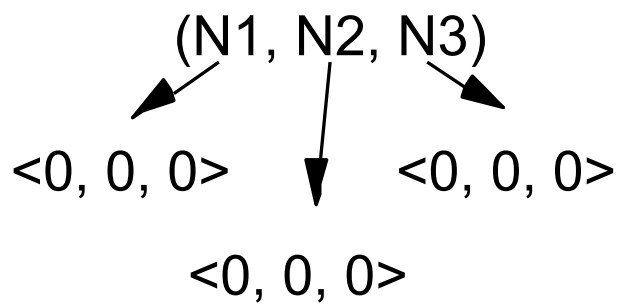




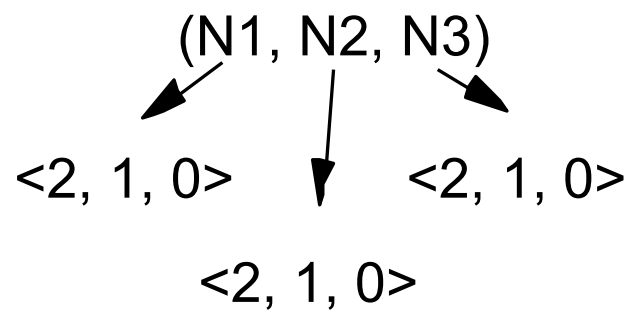
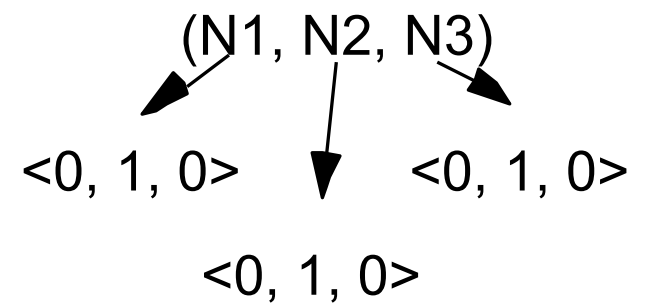
N2 updates
once



network connected.



N2 updates
once



N1 updates
twice

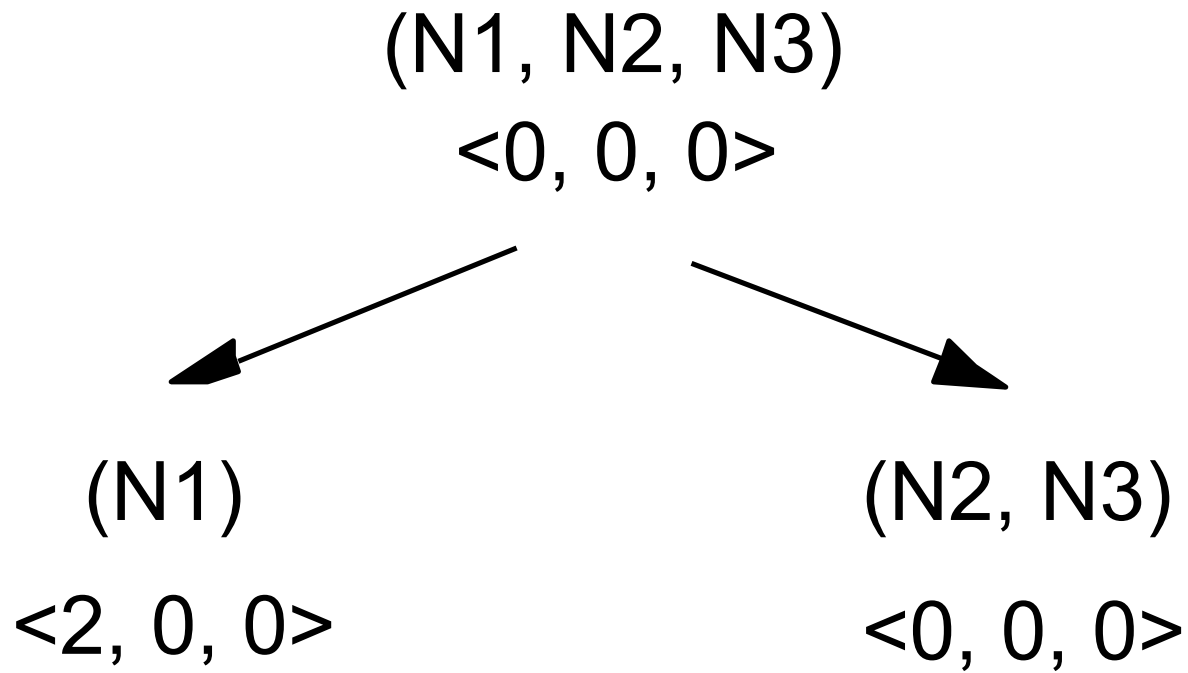
network connected

Network partition^{分区}

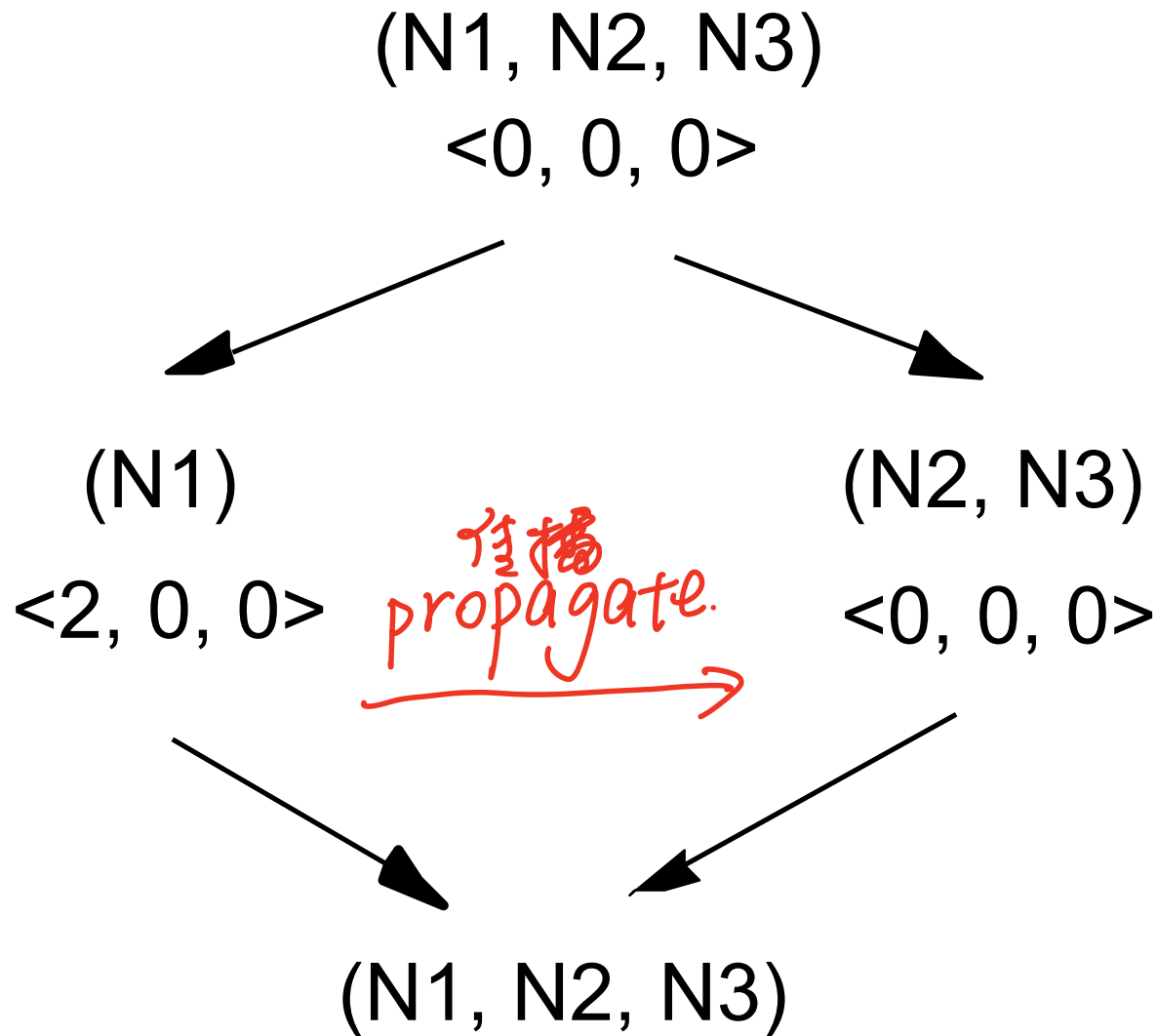
(N1, N2, N3)

<0, 0, 0>

网不通



split
N1 updates
twice



split
N1 updates
twice

repair net work
merge

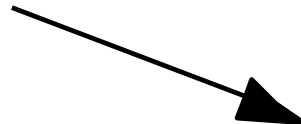
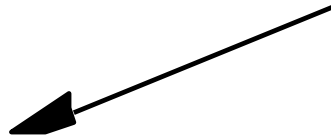
$(N1, N2, N3)$

$\langle 0, 0, 0 \rangle$

分正状态

(N1, N2, N3)

$\langle 0, 0, 0 \rangle$



(N1)

$\langle 2, 0, 0 \rangle$

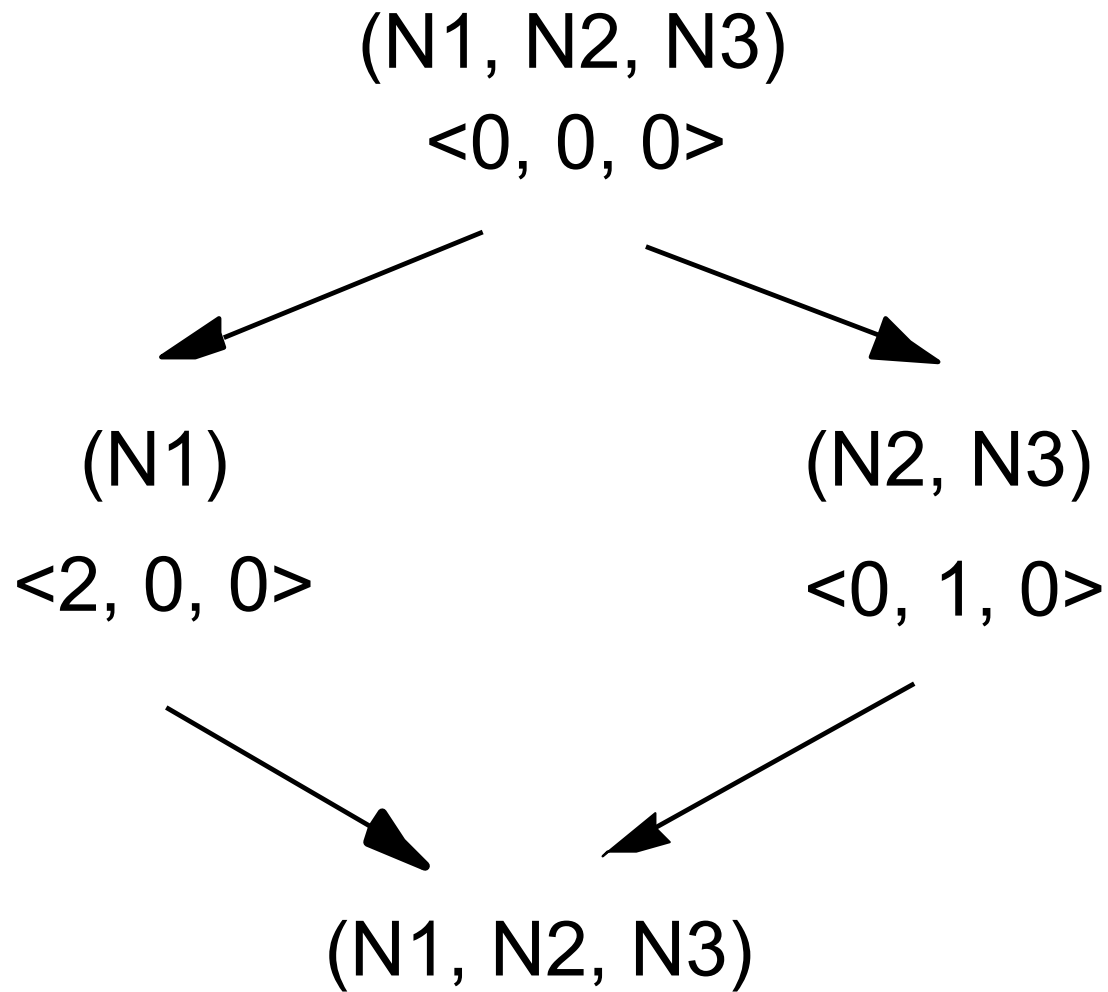
(N2, N3)

$\langle 0, 1, 0 \rangle$

split

N1 updates twice

N2 updates once



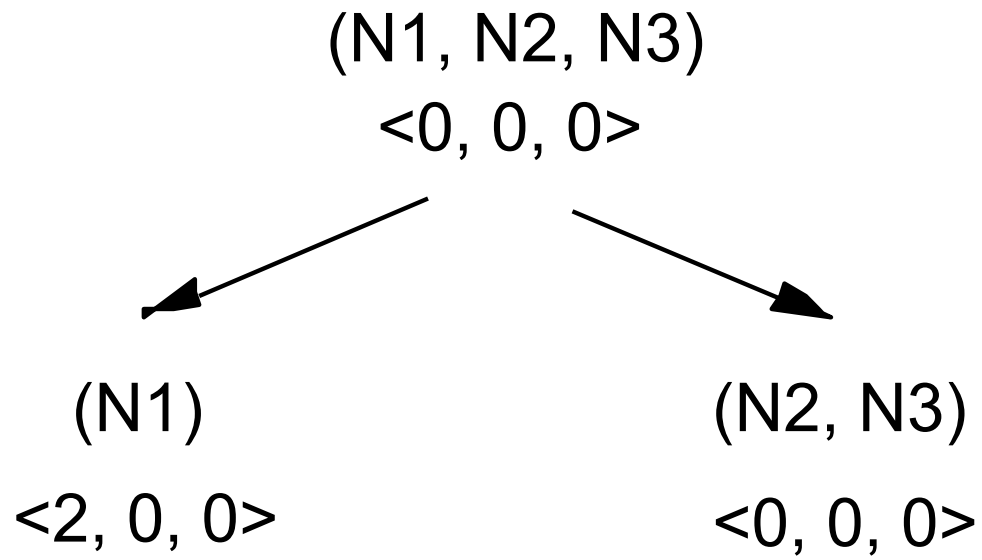
split
N1 updates twice
N2 updates once

merge

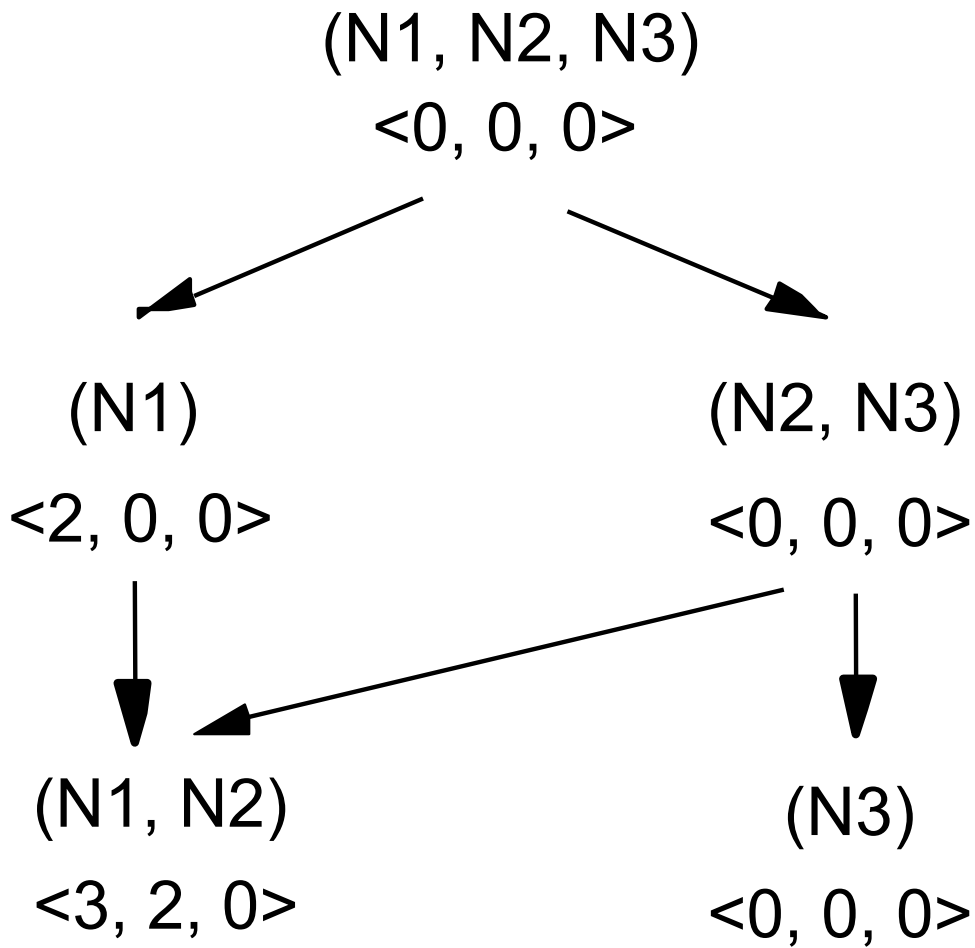
相对应的值大于等于可说
兼容 $\rightarrow$ 一致性

- Two vectors, $\langle V11, V12, \dots, V1n \rangle$ and $\langle V21, V22, \dots, V2n \rangle$, are called **compatible**, if $\langle V11, V12, \dots, V1n \rangle \geq \langle V21, V22, \dots, V2n \rangle$ or $\langle V21, V22, \dots, V2n \rangle \geq \langle V11, V12, \dots, V1n \rangle$. Otherwise, the two vectors are incompatible.
- The mutual **inconsistency** exists only when the vectors from different sites are not compatible.

$(N1, N2, N3)$
 $\langle 0, 0, 0 \rangle$



split
N1 updates
twice



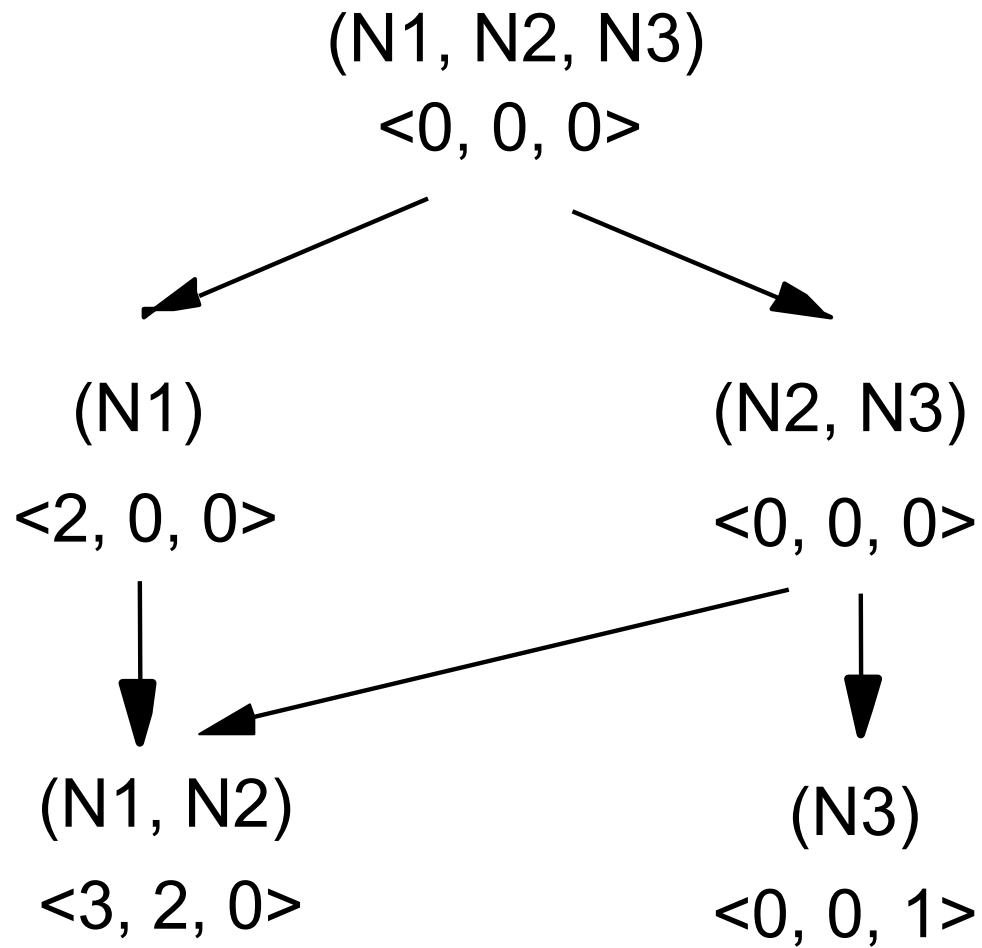
split
N1 updates
twice

sub network 1
✓ 2

split and merge
N1 updates once
N2 updates twice

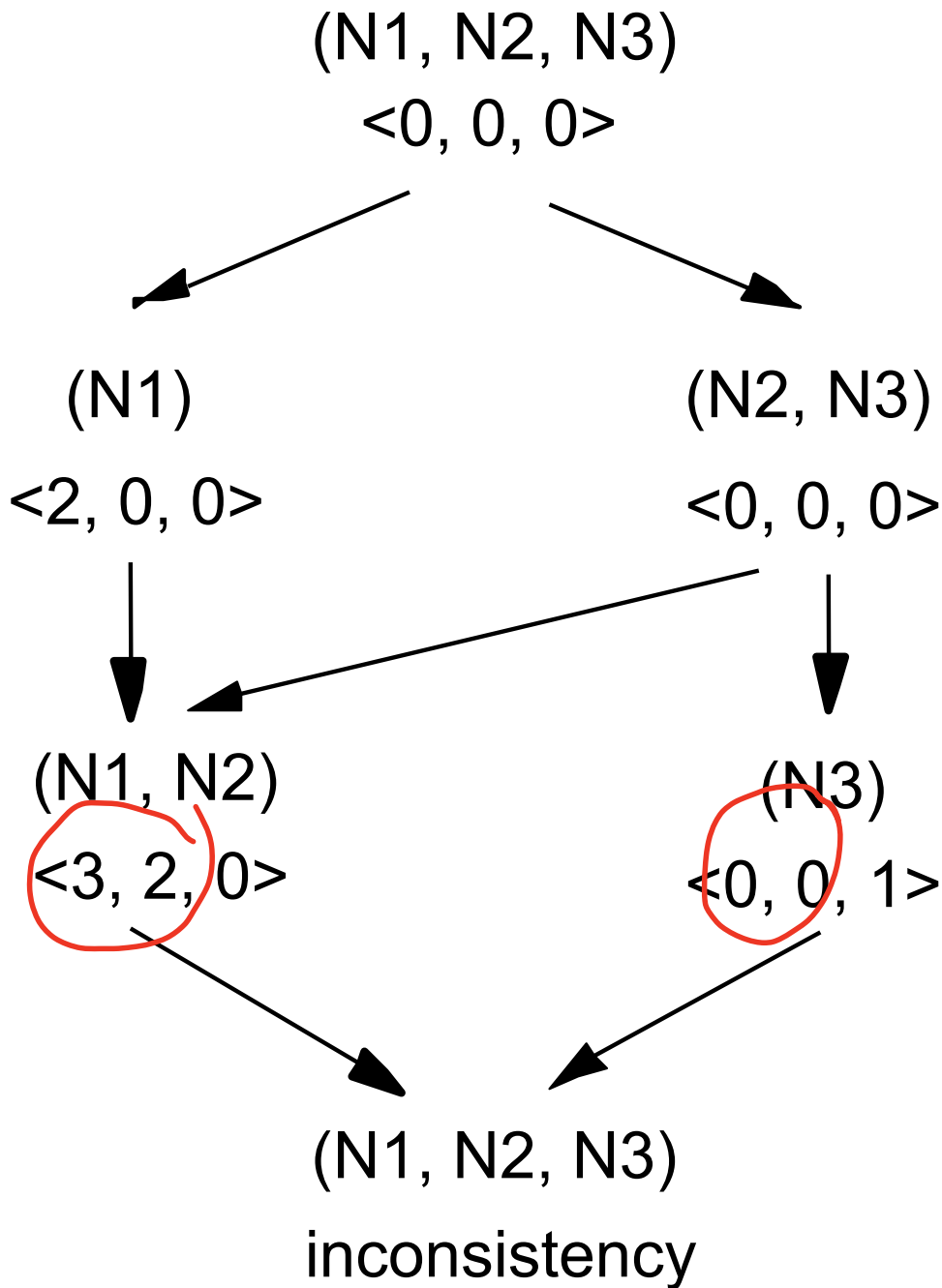
connected

3 lost



split
N1 updates
twice

split and merge
N1 updates once
N2 updates twice
N3 updates once



split
N1 updates
twice

split and merge
N1 updates once
N2 updates twice
N3 updates once

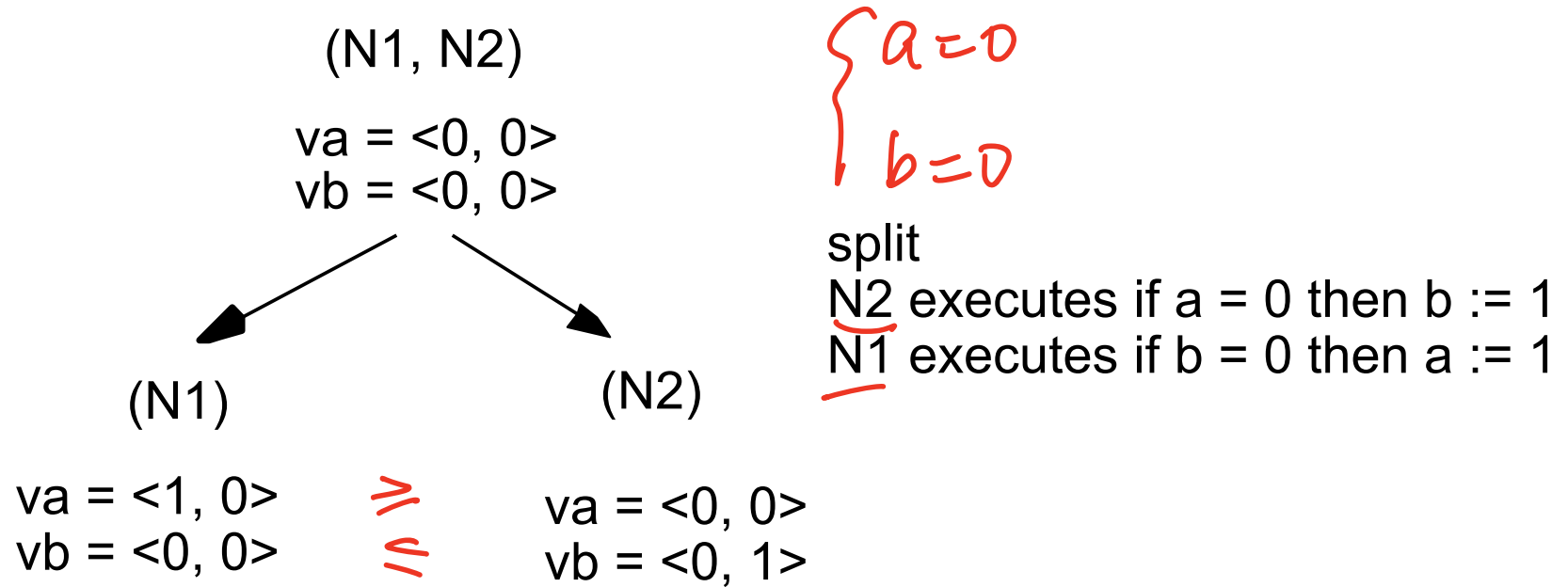
The Limitation of the Algorithm

$(N1, N2)$

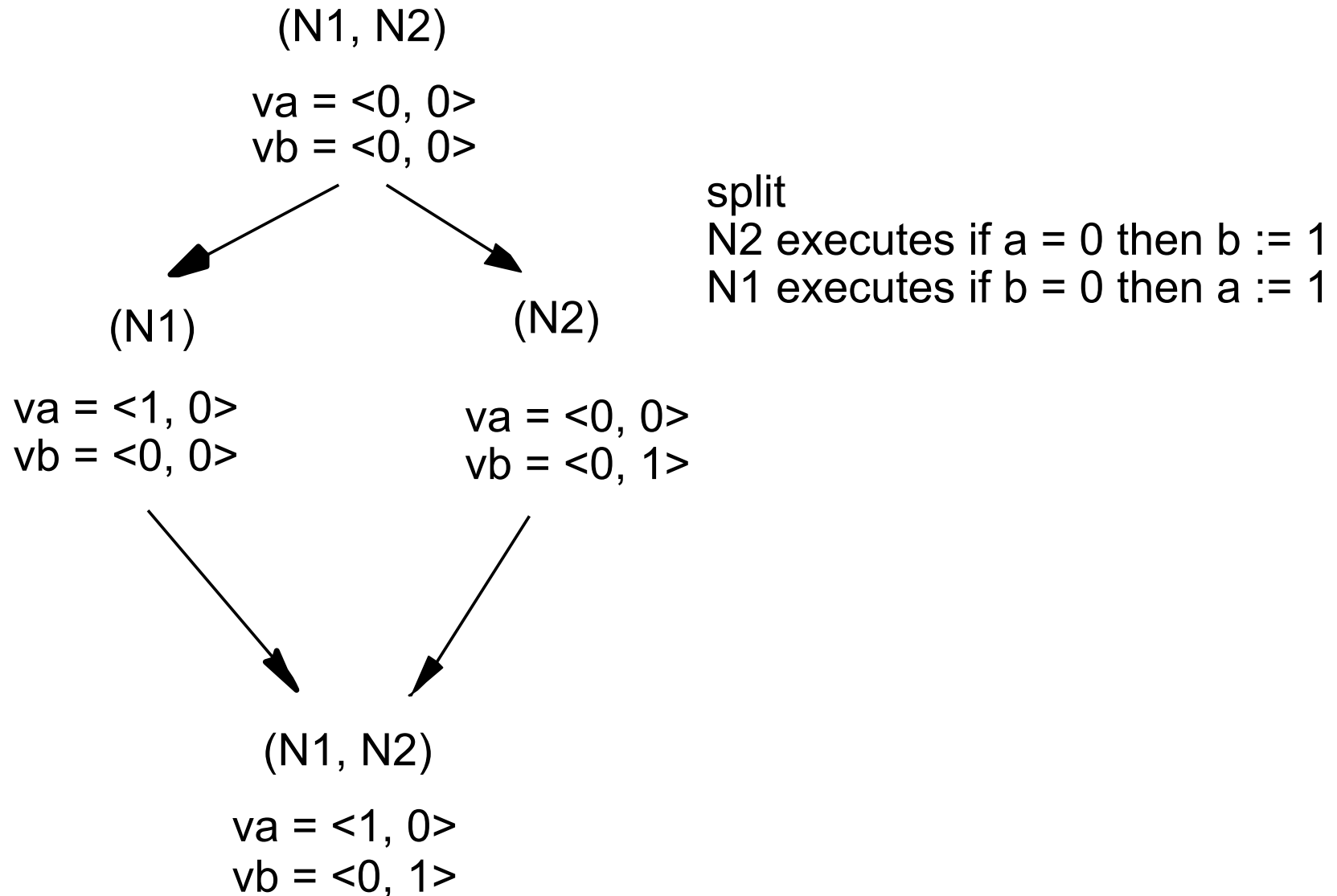
$va = \langle 0, 0 \rangle$

$vb = \langle 0, 0 \rangle$

The Limitation of the Algorithm



The Limitation of the Algorithm



reviews

- When replicated data have different values, it does not necessarily mean that the data are inconsistent. Explain why this is the case.
- Understand how Parker's algorithm works and the limitation of the algorithm.
- Find a solution to overcome the limitation of Parker's algorithm discussed in the lecture. Discuss the drawbacks (if any) of your solution.

Further reading

- D. Stott Parker et al. Detection of Mutual Inconsistency in Distributed Systems. IEEE Transactions on Software Engineering, 240-247, May 1983